Some applications of algebraic coding theory

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Abstract. In this talk, we present a wealth of interconnections between algebra, geom- etry, and combinatorics, and the areas of coding theory and its applications, such as quantum computation, distributed storage, matrix multiplication, and cryptography. We will see how properties of vanishing ideals, together with their properties like Grobner bases, Hilbert functions, footprints, degrees, being decreasing, or being Gorenstein; the study of curves, including Riemann-Roch space associated with special divisors; and the counting of lattice points, such as integer points inside of certain polyhedra, play roles in these applications.

Keywords. Coding theory, evaluation codes, algebraic coding theory.